Technical Service Bulletin

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Safe Lead Acid Battery Work Instructions	

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

The purpose of this document is to make all repair locations are aware of the safe lead acid battery work instructions. This document **only** applies to lead acid batteries.

Service Instructions

Battery Maintenance:

- Battery maintenance is fundamental to the battery achieving intended lifespan. Batteries **must** be inspected and tested at the manufacturer's maintenance interval or every 6 months, whichever is sooner.
- Battery life can be negatively affected by ambient air temperature and vibration. Recommended replacement interval is **not** to exceed 24 months at average temperature of 27°C [80.6°F].

Battery Inspection:

For activities beyond visual inspection, Lockout/Tagout **must** be performed to prevent unwanted cranking of engine.

A visual inspection **must** be performed before any battery testing is performed.

Perform a visual inspection on the following:

- Check leads and terminal posts for corrosion and tightness.
- The top of battery **must** be clean and dry with no "acidic" odor present.
- Check the top, sides, and bottom of battery for deformation o.r cracks
- Verify battery vent is **not** kinked or obstructed.

If deformation or cracks are discovered the battery **must** be replaced. Verify cold cranking amperes (CCA) for the engine is correct and replace with the appropriate maintenance type battery. If there are multiple batteries in the system, replace all batteries.

Battery Testing:

When refilling or testing electrolytes the following personal protective equipment (PPE) **must** be worn:

- Goggles: Provides protection from electrolyte splashes.
- Face shield: Protects facial skin area.
- Rubber gloves: Provides both acid protection and electrical resistance to prevent shocks.
- Rubber apron: Provides protection to the body

When load testing or performing high current starting events with battery covers off, use a di-electric battery blanket to cover the batteries to deflect any electrolyte/debris in the case of an unexpected battery explosion

Maintenance-Free Type Batteries

- Sealed and do not require addition of electrolyte. Maintenance free batteries are designed to capture evaporated electrolyte and allow it to condense and return to the battery cells.
- Often have a green dot or other indicator on the outside of the battery case to assist in determining the battery's state of charge.
- The green dot only measures the charge of a single cell, therefore, voltage checks must be performed with a load tester.
- Typical power generation application includes the use of a constant float charger. Extra care **must** be taken when using Maintenance-Free Type batteries in this application.
    - Since there is no way to check the level of the electrolyte, Cummins In. recommends that maintenance free batteries be replaced at no more than 24 months
    - Battery charger thermal compensation feature is highly recommended in this application

Maintenance Type Batteries

- Often referred to as conventional batteries
- Have removable filler caps to allow the electrolyte in each cell to be topped off and kept at the prescribed level. Proper PPE is required every time electrolytes are checked or filled.
- **Only** distilled water is to be used for maintaining proper fluid level.
- A refractometer or hydrometer **must** be used to test the level of charge in the batteries.

Table 1 below will assist in determining maintenance type battery state of charge based on specific gravity readings from a refractometer or hydrometer.

Table 1, Battery State of Charge by Specific Gravity

Battery State of Charge	Specific Gravity at 27°C [80°F]
100 Percent	1,260 to 1,280
75 Percent	1,230 to 1,250
50 Percent	1,200 to 1,220
25 Percent	1,170 to 1,190
Discharged	1,110 to 1,130

Testing can be completed with a battery load tester or a combination battery & electrical system tester. Ensure the device is calibrated and provides constant readings.

Safe working procedures:

- Ensure safe working procedures for battery inspection are followed during all repair and maintenance events. During start up or commissioning activities, battery door **must** remain closed.
- Where batteries are **not** contained within secured cabinets, batteries **must** be covered with an insulated battery blanket. Use of a battery blanket is recommended when working on and around any battery.

Moving Batteries:

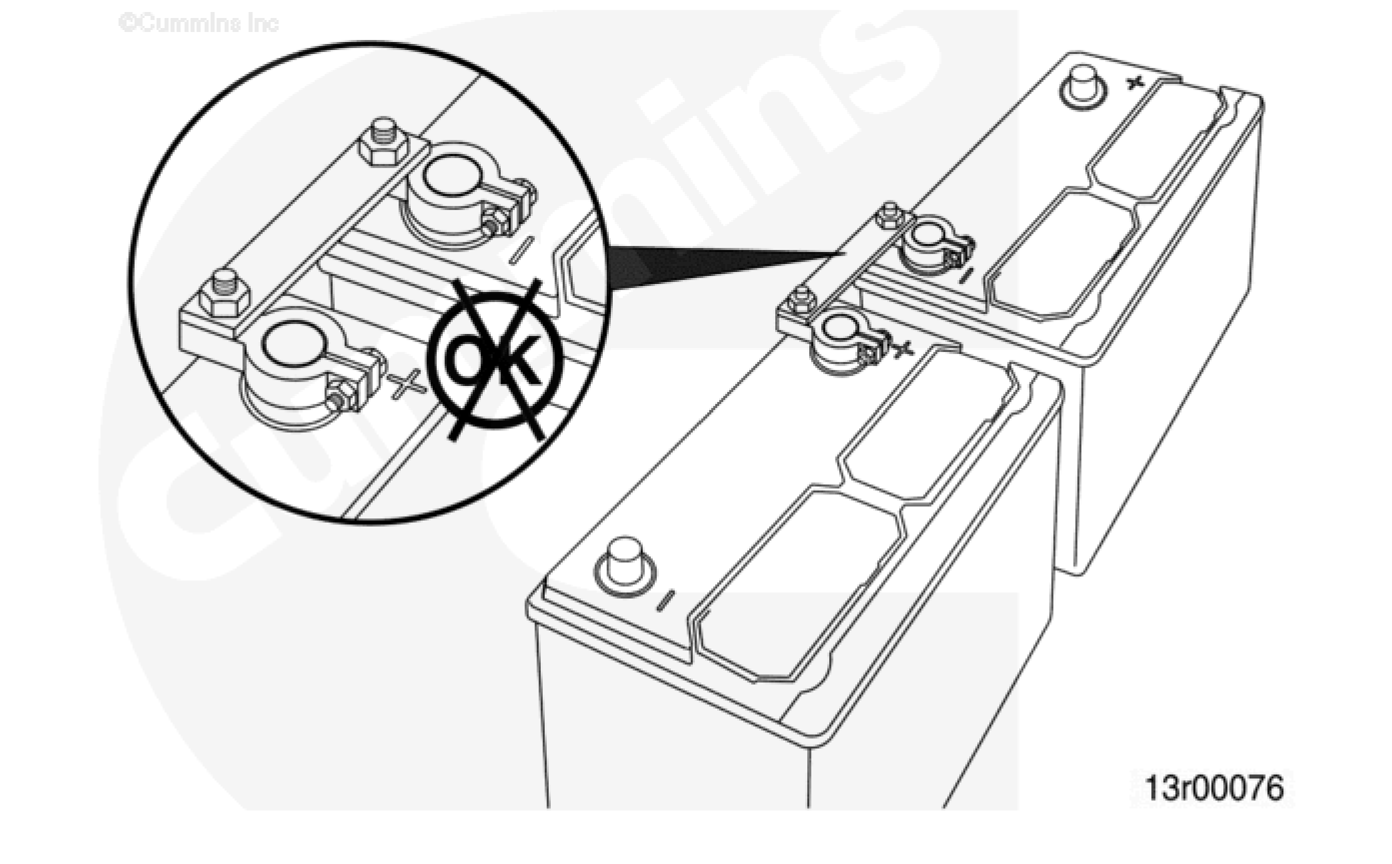
- Battery lifting over 50lbs [23kg] **must** utilize proper lifting equipment or a second technician should be used to complete the lift.
- If for any reason batteries need be moved or relocated you **must** allow 15 minutes for the battery to stabilize before you apply any load.
- Replace batteries to correct position before being placed back on load.

Battery Charging:

- Be aware of ambient conditions where battery will be installed. Ideal temperature for is 27°C [80.6°F].
- Every 8°C [14.4°F] rise in temperature decrease the life of the battery by approximately 25%.
- Use a battery charger that has thermal compensation to reduce charge voltage to an acceptable level.

Battery Connections:

- Flexible links or cable connections are recommended to allow for expansion.
- **Never** stack batteries on top of each other.
- **Always** verify proper ventilation.
- **Never** use a solid bar connection.



Emergency Response and Clean-up:

If battery acid makes contact with skin or eyes, immediately flush with water for 10 minutes and seek medical attention. If no injuries, proceed to neutralizing the acid by utilizing a battery acid spill kit. Proper PPE **must** continue to be worn when performing this task

Actions:

- Wear appropriate PPE
- Perform regular inspections and maintenance on battery
- Ensure charging voltage is within specifications of the manufacturer
- Ensure all parts and service staff involved in supporting this product, are fully aware of the above mentioned description.
- Take notice of your conditions where the battery is positioned and also the environment.
- Other battery types **not** mentioned above, should be inspected and replaced per the manufacturer's guidelines.

Technical Assistance: Call the local Cummins® Distributor or health, safety, and environment (HSE) leader for assistance.

Document History

Date	Details
2018-4-12	Module Created
2018-5-17	Corrected temperature degree conversion error in Battery Charging section.

Date	Details
2020-2-21	Updated with statement about di-electric battery blanket.
2022-5-25	Updated title to clarify lead acid battery type.
2024-10-11	No changes made to TSB, just checking TSB contents.

Last Modified: 11-Oct-2024